

Home Untitled1

localhost:8888/notebooks/Untitled1.ipynb

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JupyterLab Python 3 (ipykernel)

```
[notice] A new release of pip is available: 23.2 -> 24.0.1
[notice] To update, run: C:\Users\rmit\AppData\Local\Python\Pythoncore\3.14-64\python.exe -m pip install --upgrade pip
```

```
[5]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split, cross_val_score
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import RandomForestClassifier, GradientBoostingClassifier
from sklearn.preprocessing import LabelEncoder
from sklearn.metrics import classification_report, accuracy_score
```

```
[5]: df = pd.read_csv("C:\Users\rmit\Downloads\german_credit_data.csv")

print(df.head())
print(df.info())
```

```
Unnamed: 0  Age  Sex  Job  Housing  Saving accounts  Checking account \
0  0  67  male  2  own  little  little
1  1  22  female  2  own  little  moderate
2  2  49  male  1  own  little  high
3  3  45  male  2  free  little  little
4  4  53  male  2  free  little  little

Credit amount  Duration  Purpose
0  1169  6  radio/TV
1  9951  48  radio/TV
2  2896  12  education
3  7882  42  Furniture/equipment
4  4870  24  car

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1000 entries, 0 to 999
Data columns (total 10 columns):
 #  Column  Non-Null Count  Dtype
---  --
0  Unnamed: 0  1000 non-null  int64
1  Age  1000 non-null  int64
2  Sex  1000 non-null  object
3  Job  1000 non-null  int64
4  Housing  1000 non-null  object
5  Saving accounts  817 non-null  object
6  Checking account  686 non-null  object
7  Credit amount  1000 non-null  int64
8  Duration  1000 non-null  int64
9  Purpose  1000 non-null  object
dtypes: int64(5), object(5)
memory usage: 78.3+ KB
```

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JupyterLab Python 3 (ipykernel)

```
0  Unnamed: 0  1000 non-null  int64
1  Age  1000 non-null  int64
2  Sex  1000 non-null  object
3  Job  1000 non-null  int64
4  Housing  1000 non-null  object
5  Saving accounts  817 non-null  object
6  Checking account  686 non-null  object
7  Credit amount  1000 non-null  int64
8  Duration  1000 non-null  int64
9  Purpose  1000 non-null  object
dtypes: int64(5), object(5)
memory usage: 78.3+ KB
None
```

```
[6]: df.drop(columns="Unnamed: 0", inplace=True)
```

```
[7]: df["Saving accounts"].fillna("unknown", inplace=True)
df["Checking account"].fillna("unknown", inplace=True)
```

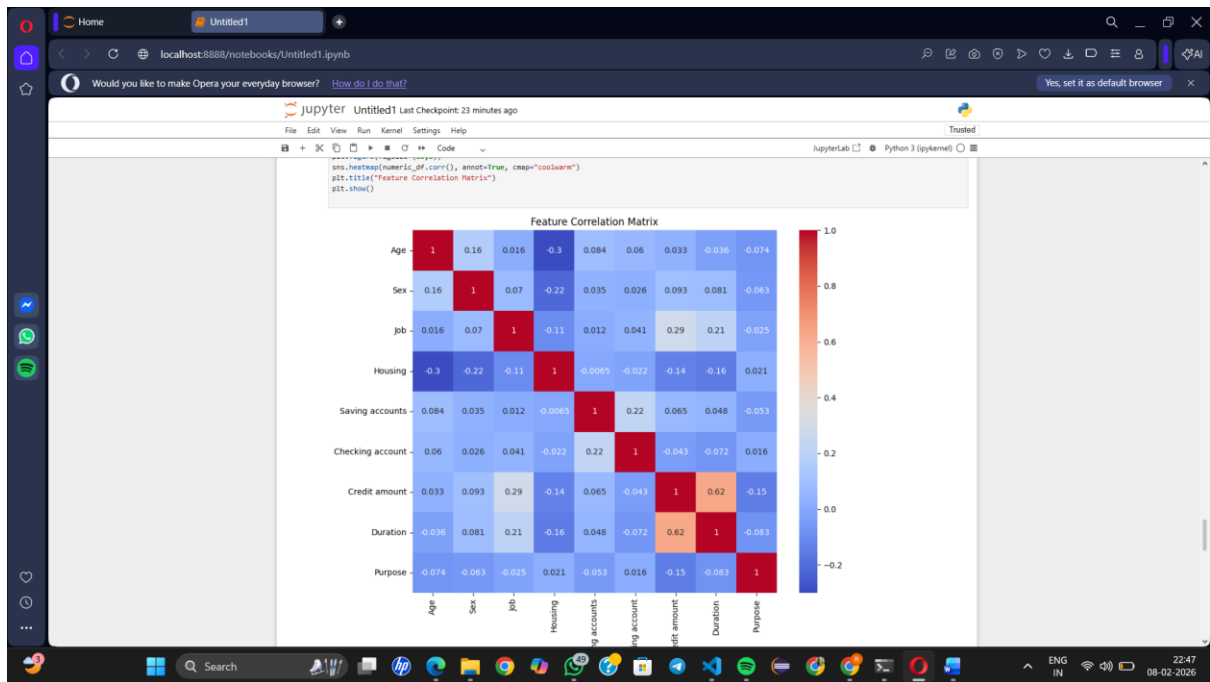
```
[8]: df["Risk"] = pd.cut(
    df["Credit amount"],
    bins=[0, 3000, 7000, np.inf],
    labels=["low", "medium", "high"]
)
```

```
[9]: le = LabelEncoder()
for col in df.select_dtypes(include="object").columns:
    df[col] = le.fit_transform(df[col])
```

```
[10]: numeric_df = df.select_dtypes(include=np.number)

plt.figure(figsize=(10,8))
sns.heatmap(numeric_df.corr(), annot=True, cmap="coolwarm")
plt.title("Feature Correlation Matrix")
plt.show()
```

Feature Correlation Matrix



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```
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```

Python 3 (spykernel)

```
[13]: X = df.drop("Risk", axis=1)
      y = df["Risk"]
      X_train, X_test, y_train, y_test = train_test_split(
          X, y, test_size=0.2, random_state=42
      )

[14]: dt = DecisionTreeClassifier()
      dt.fit(X_train, y_train)
      y_pred_dt = dt.predict(X_test)
      print("Decision Tree Accuracy:", accuracy_score(y_test, y_pred_dt))

Decision Tree Accuracy: 1.0

[15]: rf = RandomForestClassifier()
      rf.fit(X_train, y_train)
      y_pred_rf = rf.predict(X_test)
      print("Random Forest Accuracy:", accuracy_score(y_test, y_pred_rf))

Random Forest Accuracy: 1.0

[16]: gb = GradientBoostingClassifier()
      gb.fit(X_train, y_train)
      y_pred_gb = gb.predict(X_test)
      print("Boosting Accuracy:", accuracy_score(y_test, y_pred_gb))

Boosting Accuracy: 1.0

[17]: models = {
      "Decision Tree": dt,
      "Random Forest": rf,
      "Boosting": gb
    }
```

